Exercise Sheet

Lecture 13

Metric, Normed, and Hilbert Spaces

Inner products are conjugate-linear in the first slot (physics convention): $\langle x, y\rangle = \sum_n \overline{x_n} y_n$ on ℓ^2 and $\langle f, g\rangle = \int \overline{f} g$ on L^2. A $\star$ marks a harder problem.

Metrics, norms, completeness

Exercise 13.1 (proof) Show that $d(x, y) = \|x - y\|$ satisfies the triangle inequality whenever $\|\cdot\|$ is a norm.

Exercise 13.2 (computational) Decide whether the sup norm on $C[0, 1]$ comes from an inner product, using $f = 1$ and $g(x) = x$.

Exercise 13.3 (computational) For $x = (2, -1, 2) \in \mathbb{R}^3$ compute $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$.

Exercise 13.4 (computational) For $x = (1, -2, 2, -4) \in \mathbb{R}^4$ compute $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$.

Exercise 13.5 (computational) For $x = (3i, -4) \in \mathbb{C}^2$ compute $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$.

Exercise 13.6 (computational) Does the taxicab norm $\|\cdot\|_1$ on $\mathbb{R}^2$ come from an inner product? Test the parallelogram law with $u = (1, 0)$, $v = (0, 1)$.

Exercise 13.7 (computational) Verify the parallelogram law for the Euclidean norm on $\mathbb{R}^2$ with $u = (3, 0)$, $v = (0, 4)$, consistent with $\|\cdot\|_2$ coming from an inner product.

Exercise 13.8 (computational) Is $d(x, y) = (x-y)^2$ a metric on $\mathbb{R}$? Justify with a single numerical test.

Exercise 13.9 ($\star$) Prove that a normed space is complete if and only if every absolutely convergent series converges.

Exercise 13.10 (proof) Prove the reverse triangle inequality $|\,\|u\| - \|v\|\,| \le \|u - v\|$ in any normed space.

ℓ^2 and L^2

Exercise 13.11 (computational) Is $x_n = 1/n$ in ℓ^2? Is $x_n = 1/\sqrt{n}$?

Exercise 13.12 (computational) Compute $\langle f, g\rangle$ for $f(x) = 1$, $g(x) = x^2$ on $L^2[0, 1]$, and $\|g\|$.

Exercise 13.13 (computational) Normalize $\psi(x) = e^{-x}$ on $[0, \infty)$ in L^2.

Exercise 13.14 (computational) For $x = (1, \frac{1}{2}, \frac{1}{4}, \dots) \in \ell^2$ (that is $x_n = 2^{-n}$, $n \ge 0$) compute $\|x\|$.

Exercise 13.15 (computational) For $x_n = 2^{-n}$ and $y_n = 4^{-n}$ ($n \ge 0$) in ℓ^2, compute $\langle x, y\rangle$.

Exercise 13.16 (computational) For which real r is $(1, r, r^2, r^3, \dots) \in \ell^2$, and what is its norm then?

Exercise 13.17 (computational) Compute $\langle f, g\rangle$ for $f(x) = x$, $g(x) = x^2$ on $L^2[0, 1]$.

Exercise 13.18 (computational) Normalize $\psi(x) = x^2$ on $[0, 1]$ in L^2.

Exercise 13.19 (proof) Prove that if $x = (x_n) \in \ell^2$ then $x_n \to 0$. Show by example that the converse fails.

Orthonormal bases and Fourier series

Exercise 13.20 (computational) The functions $\frac{1}{\sqrt{2\pi}}e^{inx}$ are orthonormal on $L^2[-\pi, \pi]$. Verify orthonormality for $n = 0$ and $n = 1$.

Exercise 13.21 (computational) Find the Fourier sine coefficients of $f(x) = x$ on $[-\pi, \pi]$ and state the Parseval sum they yield.

Exercise 13.22 (computational) On $L^2[-1, 1]$ show that 1 and x are orthogonal, and compute $\|1\|$ and $\|x\|$.

Exercise 13.23 (computational) Normalize 1 and x on $L^2[-1, 1]$ to obtain the orthonormal pair e_0, e_1.

Exercise 13.24 (computational) Verify orthonormality of $e_n = \frac{1}{\sqrt{2\pi}}e^{inx}$ on $L^2[-\pi, \pi]$ for the indices $n = 2, 3$: compute $\langle e_2, e_2\rangle$ and $\langle e_2, e_3\rangle$.

Exercise 13.25 (computational) Find the best approximation of $f(x) = x$ on $L^2[-\pi, \pi]$ by the orthogonal set $\{\sin x, \sin 2x\}$.

Exercise 13.26 (computational) Let $\{e_1, e_2, e_3\}$ be orthonormal in a Hilbert space and v satisfy $\|v\|^2 = 10$ with $\langle e_1, v\rangle = 1$, $\langle e_2, v\rangle = 2$, $\langle e_3, v\rangle = 2$. Compute $\|Pv\|$, where P projects onto $\operatorname{span}\{e_1, e_2, e_3\}$, and the squared distance $\|v - Pv\|^2$; check Bessel's inequality.

Exercise 13.27 (proof) For an orthonormal set $\{e_n\}$ prove Bessel's inequality $\sum_n |\langle e_n, v\rangle|^2 \le \|v\|^2$.

Exercise 13.28 (proof) For an orthonormal set $\{e_1, \ldots, e_N\}$ and scalars $c_1, \ldots, c_N$, prove the Pythagorean identity $\|\sum_{n=1}^N c_n e_n\|^2 = \sum_{n=1}^N |c_n|^2$.

Projection and Riesz

Exercise 13.29 (computational) Find the orthogonal projection of $f(x) = x$ onto the constants in $L^2[0, 1]$, and the residual.

Exercise 13.30 (computational) Find the orthogonal projection of $f(x) = x^2$ onto the constants in $L^2[0, 1]$, and the residual.

Exercise 13.31 (computational) Find the orthogonal projection of $f(x) = x^3$ onto $\operatorname{span}\{1, x\}$ in $L^2[-1, 1]$.

Exercise 13.32 (computational) By Riesz, find the vector $u \in \ell^2$ representing the bounded functional $\varphi(x) = \sum_{n\ge 1} x_n/3^n$, i.e. with $\varphi = \langle u, \cdot\rangle$.

Exercise 13.33 (computational) By Riesz, find $u \in L^2[0,1]$ with $\varphi(f) = \langle u, f \rangle$ for the functional $\varphi(f) = \int_0^1 (2+3x)\, f(x)\, dx$.

Exercise 13.34 ($\star$) The functional $\varphi(x) = \sum_n x_n/2^n$ on ℓ^2 is bounded. Find the vector $u \in \ell^2$ with $\varphi = \langle u, \cdot \rangle$.

Exercise 13.35 (proof) Show that the vector u representing a bounded functional via Riesz is unique: if $\langle u, \cdot \rangle = \langle u', \cdot \rangle$ as functionals, then $u = u'$.